The Protein Journal (2026) 45:22–38
https://doi.org/10.1007/s10930-025-10310-8

Advantages and Limitations of AlphaFold in Structural Biology: Insights from Recent Studies

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Received: 8 October 2025 / Accepted: 20 November 2025 / Published online: 1 December 2025
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Abstract
Over the past three years, AlphaFold—a deep learning–based protein structure prediction system—has transformed structural biology by providing near–experimental accuracy models directly from amino acid sequences. This narrative review synthesizes applications reported in the 2022–2025 literature across human, microbial, and viral systems, drawing on peer-reviewed studies as our data source. Representative examples include modeling of SARS-CoV-2 spike and nucleocapsid proteins in virology, assisting cryo-EM interpretation of bacterial ribosomal and membrane-protein complexes in microbiology, and refining conformational hypotheses for human GPCRs in biomedicine. Across these cases, AlphaFold predictions have complemented experimental workflows by accelerating hypothesis generation, improving model fitting within ambiguous density regions (poorly resolved areas of cryo-EM maps), and guiding mutagenesis strategies to probe dynamic conformational states. We also summarize recent method extensions: AlphaFold-Multimer improves multi-chain complex assembly prediction, while molecular dynamics (MD) simulations augment AlphaFold's static models by sampling conformational flexibility and testing stability. Despite these advances, important limitations remain—particularly for intrinsically disordered regions, protein–ligand and protein–cofactor interactions, and very large or transient assemblies—and current community benchmarks indicate that approximately one-third of residues may lack atomistic precision, underscoring uncertainty in flexible or modified segments. Framed within a clear chronological window and evidence base, our analysis highlights both the practical impact and the remaining challenges of integrating AlphaFold with experiment, outlining priorities where further methodological innovation and orthogonal validation are needed.

Keywords Cryo-Electron microscopy (cryo-EM) · Cryo-Electron tomography (cryo-ET) · AlphaFold/AlphaFold2 · Integrative structural biology · Protein–protein interactions · Subtomogram averaging · Model validation

1 Introduction

1.1 Cryo-electron Microscopy (Cryo-EM)

Cryo-electron microscopy (cryo-EM) is a powerful technique for determining the three-dimensional (3D) structures of purified biological macromolecules, including proteins and viral particles [1]. In this method, samples are rapidly frozen to preserve their native conformation and are then imaged by a transmission electron microscope using an electron beam. Multiple two-dimensional (2D) projection images are collected and computationally reconstructed into a high-resolution 3D model [2].

Unlike X-ray crystallography, cryo-EM does not require crystallization, making it particularly advantageous for studying large, dynamic, or flexible proteins that are

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difficult to analyze using other structural techniques [3–5]. While cryo-EM can achieve near-atomic resolution under favorable conditions, it requires highly purified samples and is limited in capturing interactions within the native cellular environment.

1.2 Cryo-electron Tomography (Cryo-ET)

Cryo-electron tomography (cryo-ET) extends cryo-EM to enable 3D visualization of biological specimens—from purified macromolecules to intact cells—in a near-native frozen-hydrated state. Samples are vitrified to preserve native architecture, and a series of 2D projections is acquired at incremental tilt angles. These projections are reconstructed into a 3D tomogram that reveals the spatial organization of cellular components at nanometer resolution.

Unlike single-particle cryo-EM, which requires purified samples, cryo-ET can visualize macromolecules in situ, providing insights into molecular interactions within their physiological context. Nevertheless, cryo-ET faces challenges such as radiation damage, sample thickness, and the "missing wedge problem" (loss of information from limited tilt angles). Advanced computational techniques such as subtomogram averaging—where multiple repeating structures are aligned and averaged—help overcome these issues, achieving near-atomic resolution and bridging cellular and molecular scales [6, 7]. This capability makes cryo-ET indispensable for exploring viral infection mechanisms and organelle dynamics within native environments.

1.3 AlphaFold

AlphaFold, developed by DeepMind, is a deep learning framework that predicts protein structures directly from amino acid sequences [8]. By leveraging large databases of known structures, multi-sequence alignments, and transformer-based attention networks, AlphaFold achieves near-experimental accuracy and has transformed structural biology. Its iterative refinement process integrates geometric and physical constraints, as demonstrated in CASP14, where it reached sub-ångström precision for many targets.

Recent studies have shown AlphaFold's value across biological systems—for example, modeling SARS-CoV-2 spike and nucleocapsid proteins, aiding cryo-EM fitting of bacterial ribosomal complexes, and supporting mutational analysis of human GPCRs [9–11]. These applications underscore how AlphaFold accelerates hypothesis generation and interpretation of experimental data.

However, AlphaFold also exhibits intrinsic limitations that are central to this review. It primarily predicts static conformations and struggles with intrinsically disordered regions, ligand-bound states, post-translational modifications, and large multi-component assemblies [12]. As acknowledged in the AlphaFold Database publication, approximately one-third of residues do not achieve atomistic precision. These shortcomings highlight the ongoing need for integration with experimental validation, particularly through high-resolution imaging techniques such as cryo-EM and cryo-ET.

1.4 Transitional Summary and Study Rationale

Together, cryo-EM, cryo-ET, and AlphaFold represent complementary approaches that have collectively redefined structural biology. Cryo-EM offers near-atomic resolution for purified macromolecules, cryo-ET extends structural insight to the cellular level, and AlphaFold introduces data-driven prediction that bridges experimental gaps. Importantly, the convergence of AI-based prediction and high-resolution imaging now defines a new frontier in integrative structural biology.

Recent improvements in cryo-EM resolution (approaching 1.5–2.0 Å) and AlphaFold's confidence metrics (pLDDT, pTM) have made cross-validation and model integration increasingly feasible. Indeed, a growing body of work demonstrates automated fitting and flexible refinement of AlphaFold2-predicted models into cryo-EM maps — as illustrated by DeepTracer and AlphaFold2, DomainFit and elastic-network-based flexible fitting methods [13].

In this review, we critically examine recent studies (2022–2025) that integrate AlphaFold with cryo-EM and cryo-ET workflows, emphasizing both the advances achieved and the methodological limitations that remain. We frame cryo-EM and cryo-ET as contextual tools that validate, refine, or challenge AlphaFold predictions rather than as co-equal foci. This analytical perspective aligns with the review's title—"Insights from Recent Studies"—and aims to provide a literature-driven synthesis of how AI-based modeling and experimental imaging together are reshaping modern structural biology.

2 Results

2.1 Result 1 — Prevalence, Advantages, and Limitations of AlphaFold Utilization in Structural Biology Studies

Dataset and inclusion criteria. We systematically reviewed 32 studies (2022–2025) that met the following criteria: (i) primary structural results based on cryo-EM or cryo-ET; (ii) explicit mention of AlphaFold/AlphaFold2 usage or non-usage in the modeling workflow; and (iii) sufficient metadata (organism, resolution, PDB/EMD identifiers) to

support cross-study comparison (see Methods and Table 1). Targets span Homo sapiens, Sus scrofa, diverse bacterial and eukaryotic complexes, and viral assemblies (e.g., vaccinia virus). The overall resolution window for the analyzed datasets is 3.0 Å to ~ 31.7 Å, the latter reflecting inclusion of in situ cryo-ET assemblies; this clarifies the broader range reported in the Results versus the high-resolution subset summarized in Methods (3.0–4.0 Å; 9 entries).

Prevalence. 78.1% (25/32) of studies explicitly incorporated AlphaFold2 predictions, whereas 21.9% (7/32) did not (see Table 1; Supplementary Fig. S1A–B). We explicitly acknowledge that this corpus is illustrative rather than exhaustive and may under-sample the rapidly growing literature in 2024–2025.

Use-case synthesis across studies. Across AlphaFold-utilizing studies, three recurrent application modes emerged: Initial model generation for locally unresolved or medium-resolution regions of large complexes (e.g., 7–20 Å), providing atomic starting models subsequently refined against density; Domain-level fitting/validation in which high-confidence (pLDDT-high) domains were docked into maps to complete subunit assignment and define domain boundaries and inter-domain orientations;Flexible refinement or interface analysis to rationalize protein–protein interfaces and oligomeric arrangements when cryo-EM/cryo-ET maps alone were insufficient.Representative instances are cited in Section Result 2 (PPIs) and in Table 1 (per-study mapping of use cases).

Non-usage and rationale. By contrast, 7/32 studies did not employ AlphaFold2, typically because: (i) near-atomic maps (~3.0 Å) enabled confident de novo building; (ii) validated homology templates sufficed; or (iii) the biological focus was on non-protein components (lipids, glycans, RNA) where AlphaFold2 is currently inapplicable (see Table S2 for study-level details).

Limitations observed. Consistent with prior reports, performance declines for intrinsically disordered regions (IDRs), multi-state/heterogeneous assemblies, ligand-induced conformational changes, and protein–nucleic acid interactions. In such contexts, sole reliance on

Table 1 Summary of EMDB entries and AlphaFold utilization in selected structural biology studies

No.	EMD accession	Resolution (Å)	AlphaFold used	DOI link
1	EMD-40476	21.7	No	https://doi.org/10.1073/pnas.2311883121
2	EMD-18830	24.0	No	https://doi.org/10.1073/pnas.2404542121
3	EMD-18806	26.8	Yes	https://doi.org/10.1038/s44318-023-00023-y
4	EMD-45591	7.8	No	https://doi.org/10.1101/2024.06.18.599606
5	EMD-19562	20.2	Yes	https://doi.org/10.1038/s41594-024-01261-2
6	EMD-19459	16.3	Yes	https://doi.org/10.1038/s41556-025-01620-1
7	EMD-37002	31.7	No	https://doi.org/10.1038/s44318-025-00407-2
8	EMD-42509	8.9	Yes	https://doi.org/10.1016/S0092-8674(24)00973-5
9	EMD-38163	12.0	Yes	https://doi.org/10.1093/procel/pwae044
10	EMD-38162	13.0	Yes	https://doi.org/10.1093/procel/pwae044
11	EMD-38164	10.2	Yes	https://doi.org/10.1093/procel/pwae044
12	EMD-37966	9.7	Yes	https://doi.org/10.1038/s41467-024-52792-7
13	EMD-37967	7.9	Yes	https://doi.org/10.1038/s41467-024-52792-7
14	EMD-37968	12.2	Yes	https://doi.org/10.1038/s41467-024-52792-7
15	EMD-17704	7.7	Yes	https://doi.org/10.1038/s41594-024-01218-5
16	EMD-17708	8.5	Yes	https://doi.org/10.1038/s41594-024-01218-5
17	EMD-17753	13.4	Yes	https://doi.org/10.1038/s41594-024-01218-5
18	EMD-18917	30.7	Yes	https://doi.org/10.1128/mbio.03134-23
19	EMD-18918	9.7	Yes	https://doi.org/10.1128/mbio.03134-23
20	EMD-17132	3.0	Yes	https://doi.org/10.1038/s41594-024-01441-0
21	EMD-17133	3.0	Yes	https://doi.org/10.1038/s41594-024-01441-0
22	EMD-17135	3.0	Yes	https://doi.org/10.1038/s41594-024-01441-0
23	EMD-17136	3.0	Yes	https://doi.org/10.1038/s41594-024-01441-0
24	EMD-17137	3.0	Yes	https://doi.org/10.1038/s41594-024-01441-0
25	EMD-50672	3.0	Yes	https://doi.org/10.1002/2211-5463.13873
26	EMD-19136	3.0	Yes	https://doi.org/10.1126/sciadv.adk6285
27	EMD-37727	3.0	Yes	https://doi.org/10.1016/S0969-2126(24)00142-4
28	EMD-43286	3.0	Yes	https://doi.org/10.1101/2023.05.31.542975
29	EMD-19599	18.0	No	https://doi.org/10.1016/j.str.2024.04.016
30	EMD-43912	12.8	No	https://doi.org/10.1038/s41467-024-53082-y
31	EMD-46612	18.7	Yes	https://doi.org/10.3390/v16091447
32	EMD-19137	30.0	Yes	https://doi.org/10.1016/j.molcel.2024.11.038

Summary of EMDB entries analyzed for AlphaFold utilization in structural biology studies

This table compiles 32 cryo-EM and cryo-ET structures deposited in the Electron Microscopy Data Bank (EMDB) between [insert date range], which were surveyed to evaluate the prevalence, advantages, and limitations of AlphaFold usage in structural interpretation. For each entry, the EMDB accession number (EMD-XXXX), reported resolution, direct EMDB link, AlphaFold usage status, and associated DOI link to the corresponding publication are provided. AlphaFold usage was determined based on explicit mention in the methods or results sections of the cited articles. Resolutions are reported as in the EMDB metadata, and links are given in their original published format to facilitate retrieval and citation

AlphaFold-only models increased the risk of misassignment or over-interpretation, underscoring the need for orthogonal validation and density-guided refinement.

Conclusion. In aggregate, the dataset supports viewing AlphaFold2 as an integrative tool that complements cryo-EM/cryo-ET—most effectively when predictions are critically evaluated and experimentally validated, particularly in dynamic or heterogeneous systems (Table 1).

2.2 Statistical Analysis of AlphaFold Usage by Resolution

Datasets were binned into high-resolution (≤4.0 Å) versus lower-resolution (>4.0 Å) groups. Statistical tests—including chi-square ($\chi^2=0.20$, $p=0.656$), Fisher's exact ($p=0.640$), and Spearman's ρ ($\rho=-0.163$, $p=0.373$)—revealed no significant association between nominal map resolution and AlphaFold usage (Supplementary Fig. S1A–B).

Interpretation. The absence of correlation likely reflects broad adoption across resolution regimes: even at medium-to-low resolutions, AlphaFold2 domain models are routinely used for docking/annotation, whereas at near-atomic resolutions experimental density already enables confident building. In other words, decision to use AlphaFold2 is driven more by local interpretability and compositional complexity than by global nominal resolution. We note that power is limited by the sample size ($n=32$), and future expansions may refine these trends.

2.3 Result 2 — AlphaFold2 in Protein–Protein Interaction Studies

All case studies presented in this section involve the explicit use of AlphaFold2 as part of the structural determination workflow. Across diverse biological contexts—from chromatin remodeling complexes and viral core proteins to nuclear pore components and RNA virus assemblies—AlphaFold2 has been employed to generate high-confidence structural predictions that complement experimental cryo-EM and cryo-ET data, thereby resolving binding interfaces, oligomeric states, and dynamic architectures that were previously inaccessible.

2.4 Fundamental Principles and Advances in Protein–Protein Interaction Studies

Protein–protein interactions (PPIs) are fundamental to virtually all biological processes, including cellular signaling, metabolic regulation, immune defense, and structural organization. These interactions frequently occur within large and dynamic macromolecular assemblies, where subtle conformational changes can determine functional outcomes. Traditional experimental methods—such as X-ray crystallography, NMR spectroscopy, and cryo-EM—have provided invaluable insights but are often constrained by resolution, sample requirements, or their inability to capture transient states.

The advent of deep learning–based prediction tools, particularly **AlphaFold2** [10], has transformed computational structural biology by enabling high-accuracy modeling of tertiary and quaternary structures. In its "Complex Prediction" mode, AlphaFold2 can model binding interfaces and oligomeric states de novo [10].

Representative examples are shown in Fig. 1. The **E3 ubiquitin ligase–substrate complex (EMD-38163)** [11] was resolved by cryo-EM at 12.0 Å (Fig. 1A, yellow). AlphaFold2 predictions were incorporated into poorly resolved regions, revealing a hydrophobic–electrostatic interface that was later validated by mutagenesis. Similarly, the **chromatin remodeling complex (EMD-37966)** [12] was reconstructed at 9.7 Å (Fig. 1B, cyan). AlphaFold2-predicted subunits were fitted into the density to complete subunit assignments and reveal allosteric coupling between distal domains—mechanistic insights that standalone cryo-EM could not provide [1].

2.5 Structural Elucidation of Viral Core Proteins

Viruses rely on tightly regulated PPIs to assemble infectious particles and stabilize their genomes. The **vaccinia virus** provides a representative case (Fig. 2). In a landmark study, AlphaFold2 predicted the structures of five core proteins (A3, A4, A10, L4, and F17) with high confidence (pLDDT>90). These models were docked into cryo-EM maps at 7.7–13.4 Å resolution (EMD-17704, EMD-17708, EMD-17753) [13], revealing that A10 and A4 form a curved heterodimeric scaffold, while L4 and F17 stabilize the genomic core—a finding that aligned with long-standing genetic hypothese [14].

Follow-up studies extended these insights to the **immature viral core (EMD-18917, EMD-18918)** [15] reconstructed at ~30 Å (Fig. 2D–E). AlphaFold2 predictions revealed pore-like openings in the A10–A4 scaffold that may facilitate DNA packaging. Cryo-EM reconstructions showed these pores are occluded during maturation, suggesting a gating mechanism critical for viral infectivity [16].

2.6 Nuclear Pore Dynamics and Viral Antagonism

Beyond viral assemblies, AlphaFold2 has been instrumental in investigating **host–pathogen interactions**. Nuclear pore complexes (NPCs), which regulate nucleocytoplasmic transport, are frequent viral targets. In **EMD-19137** [17],

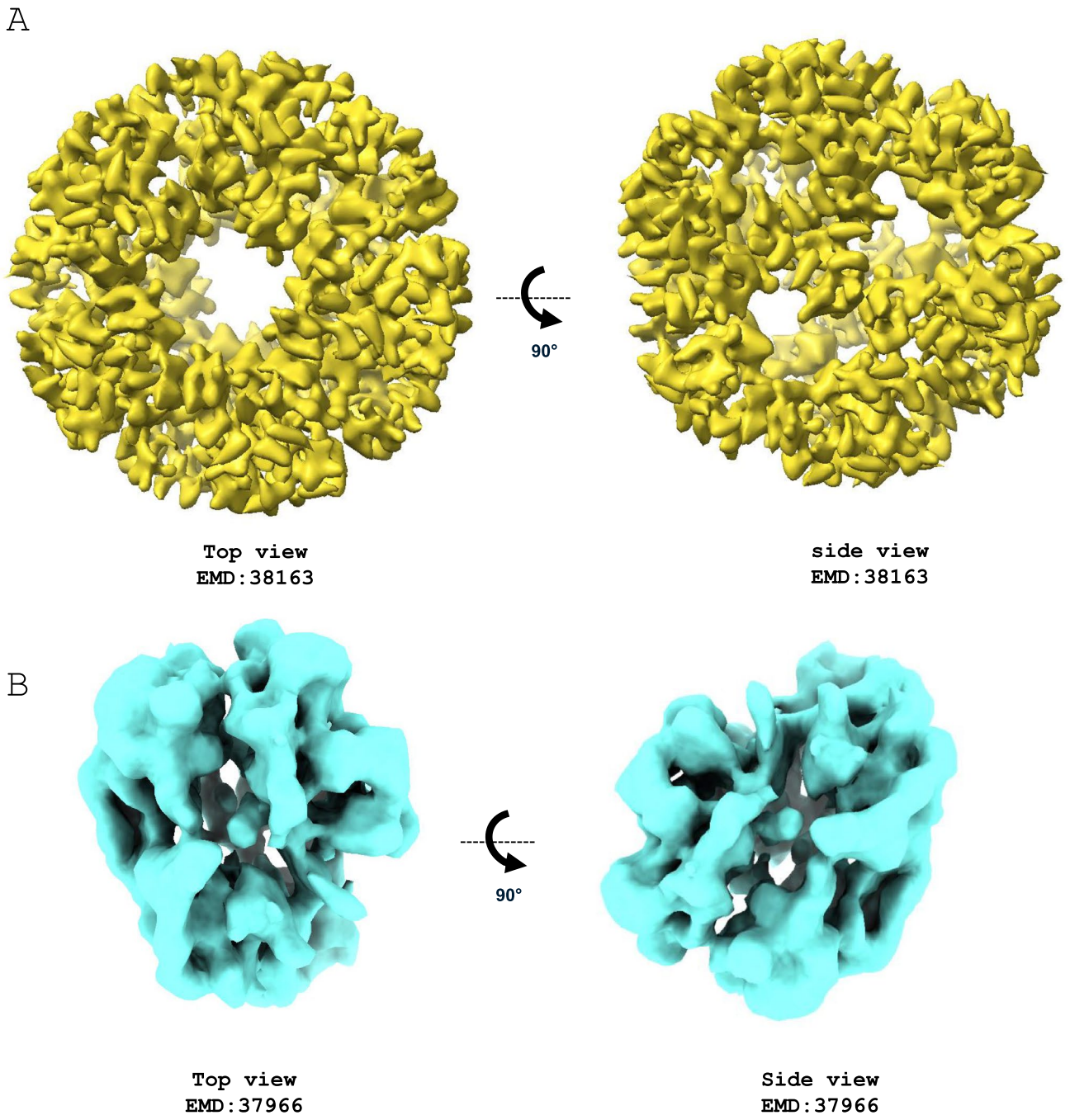

Fig. 1 Cryo-EM reconstructions of representative macromolecular complexes analyzed in this study. **A E3 ubiquitin ligase–substrate complex** (EMD-38163) resolved by single-particle cryo-EM at 12.0 Å resolution, shown in **yellow**. Left: top view; right: side view after 90° rotation. AlphaFold2 predictions were incorporated to model flexible or poorly resolved regions, revealing a hydrophobic–electrostatic interface later validated by site-directed mutagenesis. **B Chromatin remodeling complex** (EMD-37966) resolved by single-particle cryo-EM at 9.7 Å resolution, shown in **cyan**. Left: top view; right: side view after 90° rotation. AlphaFold2-predicted subunit structures were fitted into the density map to complete subunit assignments and identify allosteric coupling between distal domains, offering mechanistic insights into chromatin remodeling. Maps were obtained from the Electron Microscopy Data Bank (EMDB) and visualized using UCSF Chimera

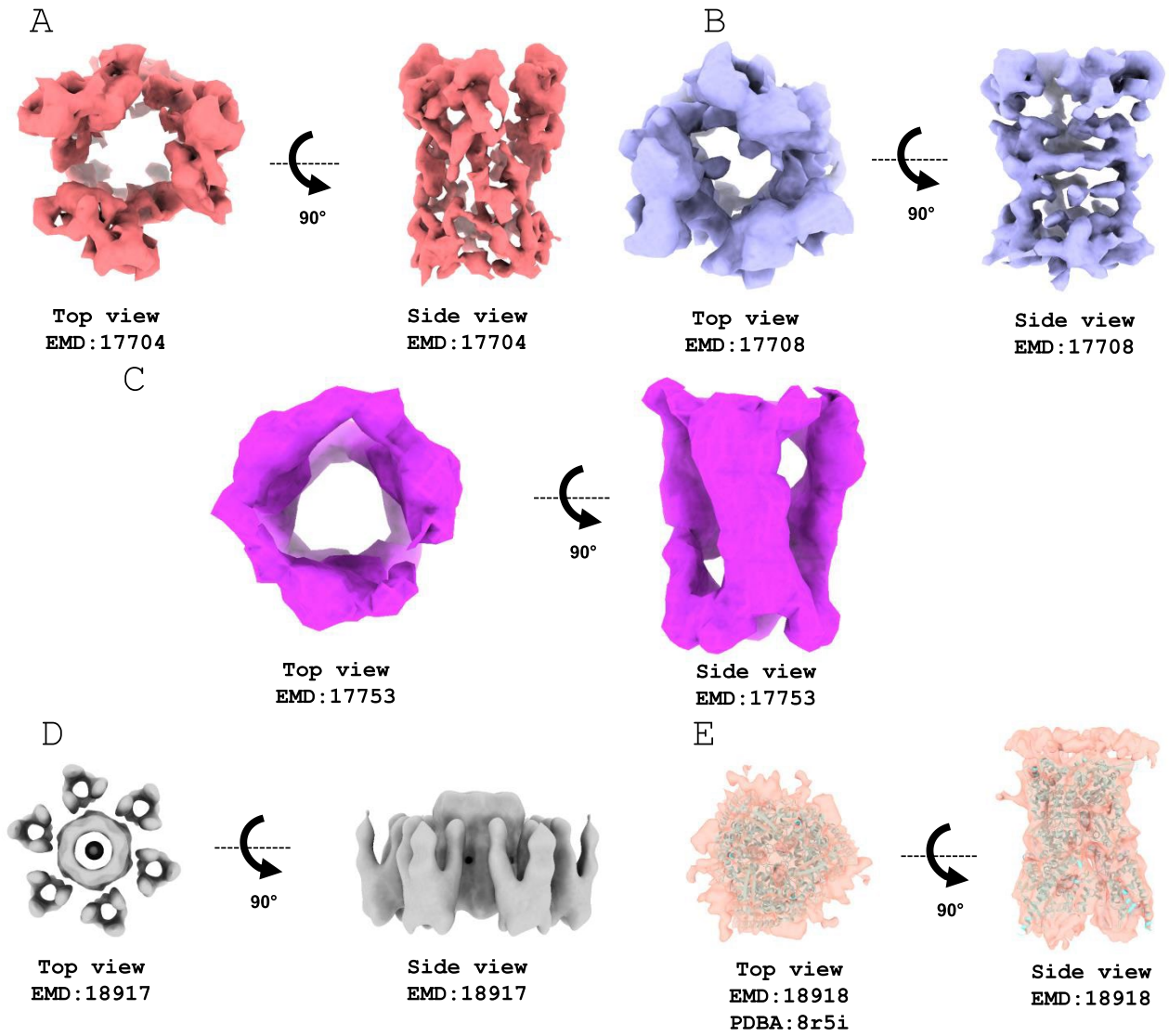

Fig. 2 Cryo-EM reconstructions of vaccinia virus core proteins and immature core architecture. **A A3 core protein** (EMD-17704) resolved by single-particle cryo-EM at 7.7 Å resolution, shown in **red**. Left: top view; right: side view after 90° rotation. AlphaFold2-predicted models of A3 were fitted into the density to define its curved scaffold within the core shell. **B A4 core protein** (EMD-17708) resolved by single-particle cryo-EM at 8.5 Å resolution, shown in **light blue**. AlphaFold2 modeling revealed its heterodimeric interface with A10, contributing to capsid stability. **C L4 core protein** (EMD-17753) resolved by single-particle cryo-EM at 13.4 Å resolution, shown in **magenta**. AlphaFold2 predictions were used to dock the full-length L4 structure into the density, elucidating its role in genomic core stabilization. (D) **Immature viral core** (EMD-18917) at 30.7 Å resolution, shown in **grey**. Cryo-EM reconstruction revealed pore-like openings in the A10–A4 scaffold, proposed to facilitate DNA packaging; AlphaFold2 provided subunit models for fitting. (E) **Immature viral core with fitted atomic model** (EMD-18918; PDB: 8r5i) at 9.7 Å resolution, shown in **salmon pink** with the fitted model overlaid. AlphaFold2 predictions of A10, A4, and L4 were integrated with experimental density to resolve the immature-to-mature transition mechanism. Maps were obtained from the Electron Microscopy Data Bank (EMDB) and visualized using UCSF Chimera

cryo-ET reconstructions at 30 Å (Fig. 3A, beige) revealed structural plasticity of NPCs under osmotic stress. AlphaFold2 predictions of Dictyostelium NPC components were integrated with density maps, showing reversible dilation that may support stress adaptation [18].

The same study examined **SARS-CoV-2 ORF6**, a known NPC inhibitor. AlphaFold2 predicted that ORF6 binds the cytoplasmic filaments of NPCs via a helical domain, sterically blocking nuclear import. This prediction was experimentally confirmed by fluorescence polarization assays, illustrating how AlphaFold2 can guide functional validation.

2.7 RNA Virus Assembly Mechanisms

AlphaFold2 has also provided critical insights into RNA virus organization. In **EMD-46612** [19], the **rabies virus matrix protein (M)** was reconstructed by cryo-EM at

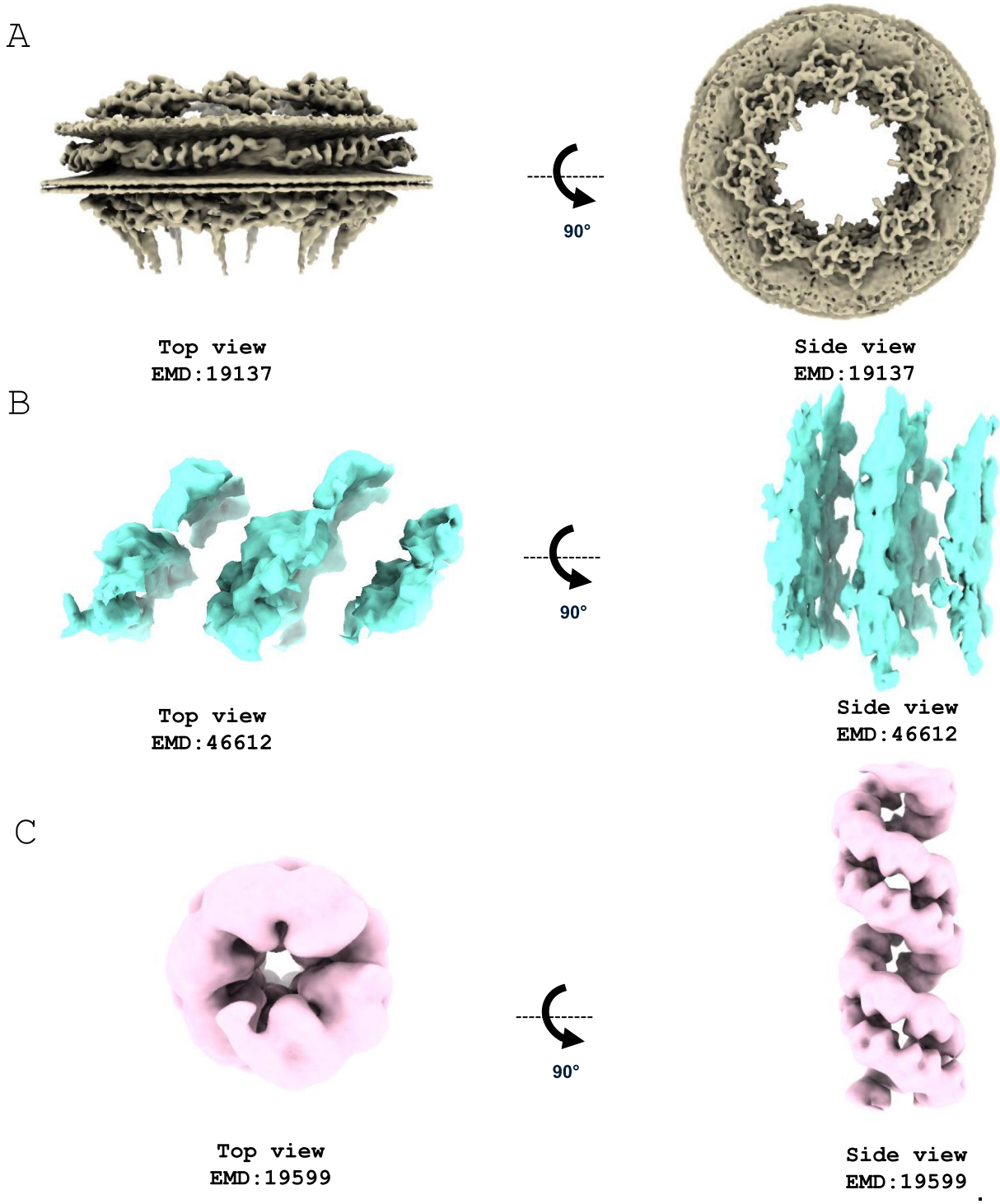

Fig. 3 Cryo-EM reconstructions of nuclear pore complexes and viral matrix/nucleoprotein assemblies. **A Nuclear pore complex (NPC) from Dictyostelium discoideum** (EMD-19137) resolved by cryo-electron tomography at 30.0 Å resolution, shown in **beige**. Left: top view; right: side view after 90° rotation. AlphaFold2-predicted models of NPC components were integrated with cryo-ET density to reveal conformational changes during osmotic stress and the binding interface of SARS-CoV-2 ORF6 with cytoplasmic filaments. **B Rabies virus matrix protein (M) bound to nucleoprotein (N)** (EMD-46612) resolved by single-particle cryo-EM at 18.7 Å resolution, shown in **cyan**. AlphaFold2-predicted models of M and N were fitted into the density map, defining a 1:1 stoichiometry that organizes the viral ribonucleoprotein complex during budding. **C Thogoto virus nucleoprotein (NP)** (EMD-19599) resolved by cryo-electron tomography at 18.0 Å resolution, shown in **pink**. AlphaFold2 modeling revealed NP as a hexameric ring, which stacks into left-handed double helices, suggesting a conserved genome-packaging mechanism among orthomyxoviruses. Maps were obtained from the Electron Microscopy Data Bank (EMDB) and visualized using UCSF Chimera

18.7 Å (Fig. 3B, cyan). AlphaFold2 predicted an elongated structure with a conserved hydrophobic groove, and when fitted into the cryo-EM density, the model revealed a 1:1 stoichiometry with nucleoprotein (N), organizing the ribonucleoprotein complex during budding [20].

Similarly, the **Thogoto virus nucleoprotein (NP)** was reconstructed by cryo-ET at 18.0 Å (EMD-19599; Fig. 3C, pink) [21]. AlphaFold2 modeling revealed NP as a hexameric ring stacking into left-handed double helices, suggesting a conserved genome-packaging mechanism across orthomyxoviruses.

2.7.1 Comparison with Alternative Modeling Approaches"

Studies comparing pipelines report that template- or fragment-based approaches (e.g., RosettaFold in low-homology membrane proteins) often fail to converge, whereas AlphaFold2 maintains plausible topology—consistent with transformer attention capturing long-range dependencies missed by template-based methods.

3 Summary

Together, these case studies highlight how AlphaFold2 complements cryo-EM and cryo-ET by providing accurate de novo models that can be docked into medium-resolution density maps. This synergy enhances model completeness, accelerates structural interpretation, and guides experimental validation. From **E3 ligase–substrate interactions** to **viral capsid assembly**, **nuclear pore plasticity**, and **RNA virus genome packaging**, AlphaFold2 has resolved previously intractable structural questions and revealed novel biological mechanisms.

3.1 Result 3. AlphaFold's Applications, Limitations, and Synergy with Cryo-EM/ET

Despite its transformative impact on single-protein modeling, AlphaFold2 remains constrained when biological insight depends on native context, mesoscale organization, or time-dependent conformational change. Reading across our dataset, the cases in this section demonstrate that cryo-EM and cryo-ET frequently provide information beyond the predictive scope of current AI models, while also showing how AlphaFold2 can still contribute as a source of high-confidence starting models that are refined against experimental density. Together, these studies delineate a practical division of labor: AlphaFold2 excels at stable domain geometry and interface hypotheses, whereas cryo-EM/ET resolves membrane association, long-range packing, and state switching in the cellular environment.

3.2 In situ macromolecular assemblies: contextual complexity beyond prediction

In situ assemblies underscore why contextual complexity exceeds sequence-to-structure prediction. In the CRP–C1 system (EMD-18830), cryo-ET visualized a four-pentamer C-reactive protein platform engaging four C1 head groups on damaged membranes, revealing an arrangement driven by surface electrostatics and curvature that cannot be inferred from isolated subunits (Fig. 4A) [22]. A similar point emerges from the LRRK2–microtubule complex (EMD-40525), where only cellular cryo-ET captured the filamentous registry and domain disposition that stabilize native filaments (Fig. 4B) [23]. Nucleosome-scale chromatin organization likewise illustrates limits of prediction: Volta phase-plate cryo-ET of RPE-1 cells (EMD-37002) showed irregular chromatin domains containing canonical mono- and dinucleosomes in G1 and densely packed, substructure-poor chromatids in metaphase; higher-order ordered fibers were absent (Fig. 4C) [24]. During mitotic condensation (EMD-19459), subtomogram averages revealed short-range nucleosome stacking rather than long-range fibers, with rearrangements linked to condensins and histone modifications (Fig. 4D) [25]. None of these mesoscale transitions are accessible to AlphaFold2 because they depend on crowding, composition, and cellular mechanics, not solely on amino-acid sequence.

3.3 Dynamic Structural Transitions and Conformational Flexibility

Dynamic structural transitions further highlight the gap between static prediction and physiological motion. In human cardiac thick filaments (EMD-40476), cryo-ET resolved three distinct myosin-head states and a regulatory role for cardiac myosin-binding protein C (cMyBP-C) in crown-level stability within sarcomeres (Fig. 5A) [26]. Because these states interchange during contraction, the relevant observables are state populations and spatial couplings across the filament, features that elude single-state AlphaFold2 models. Here, experimental mapping of state distributions provides the ground truth to which any computational dynamics (e.g., MD) must be anchored.

3.4 AlphaFold as a complementary tool: starting models and refinement

From **EMD-17132, EMD-17133, EMD-17135, EMD-17136, EMD-17137 (ZBTB20 study)** [27], researchers investigated ZBTB20-mediated transcriptional repression and its complex assembly using cryo-EM and SAXS. AlphaFold predictions were indeed employed to model

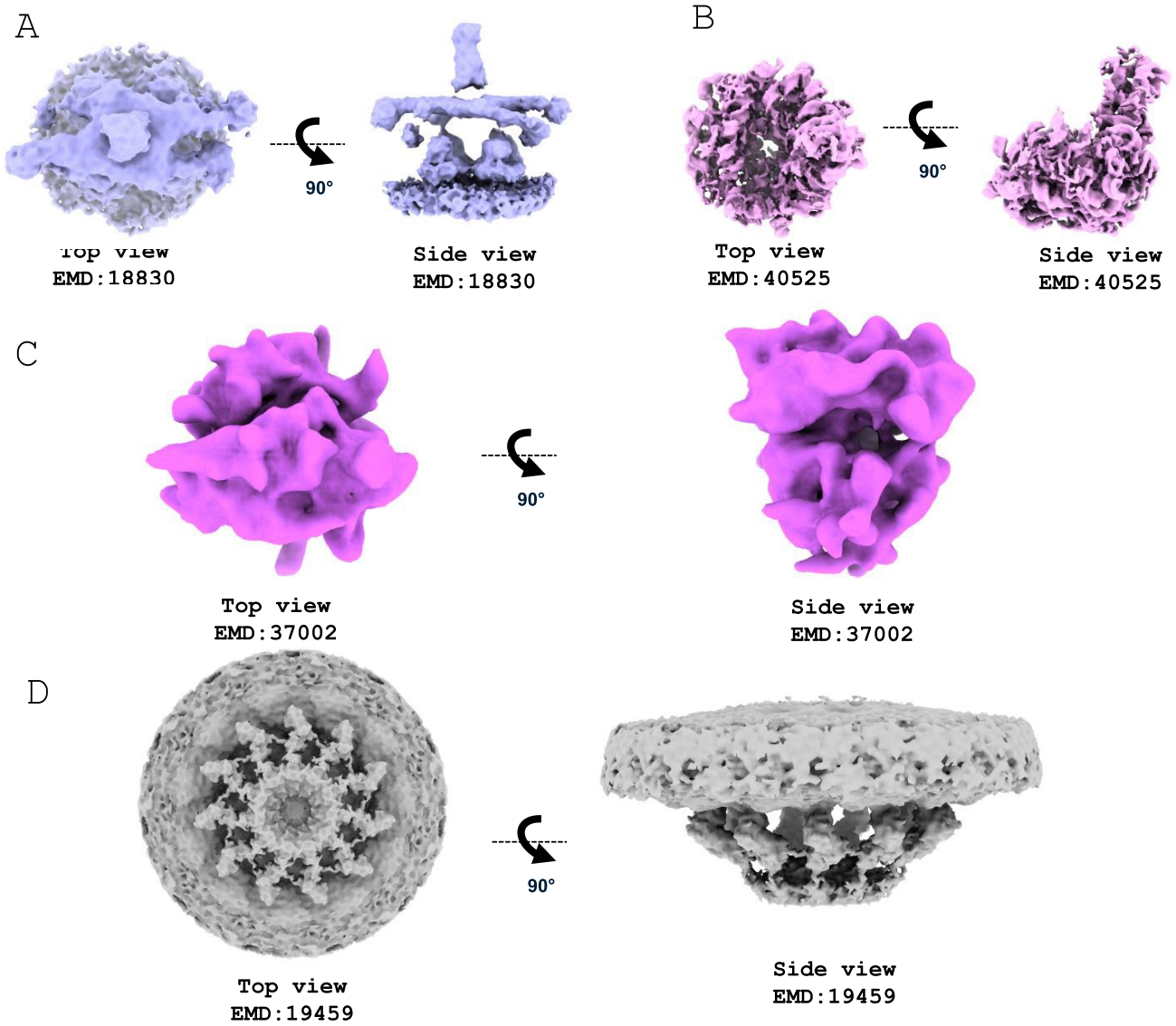

Fig. 4 Representative cryo-ET reconstructions illustrating macromolecular assemblies beyond the predictive scope of AlphaFold. **A Complement receptor platform (CRP)–C1 complex (EMD-18830):** cryo-ET revealed a four-pentamer CRP arrangement engaging four C1 head groups on damaged cell membranes, visualized in the native membrane-associated context. **B LRRK2–microtubule complex (EMD-40525):** filamentous organization and domain arrangement resolved only through in situ cryo-ET imaging, showing elongated architecture along microtubules that cannot be captured by static prediction. **C Nanoscale chromatin organization in G1 and metaphase nuclei (EMD-37002):** reconstructions revealed phase-specific chromatin compaction patterns within intact nuclei, inaccessible to AlphaFold's static predictions. **D Mitotic chromatin condensation (EMD-19459):** subtomogram averages demonstrated coordinated short-range nucleosome stacking and higher-order structural rearrangements during mitosis, emphasizing the requirement for direct experimental observation *In all cases, AlphaFold predictions were insufficient to reproduce native architecture, membrane context, or higher-order chromatin arrangements resolved by cryo-ET.*

ZBTB20's BTB domain, offering *"high-confidence initial structural insights"* particularly for segments where experimental images were unclear or represented flexible regions. However, this paper unequivocally highlighted AlphaFold's inherent limitations as a standalone solution. Its predictions consistently served as merely a *"starting model"*, necessitating rigorous validation and extensive refinement against empirical cryo-EM density maps to achieve atomic precision. The correct placement and integration of predicted structures within the intricate assembly remained critically dependent on experimental context—for instance, identifying a predicted domain solely *"on the basis of its position near the peptide exit site"* as indicated by density (Fig. 6A–E).

From **EMD-37727 (Ppp1r12c/MYPT3 study)** [28], scientists explored the structures of human and mouse ankyrin repeat domains to understand divergent RhoA-inhibitory activity. AlphaFold generated *"excellent initial models"*

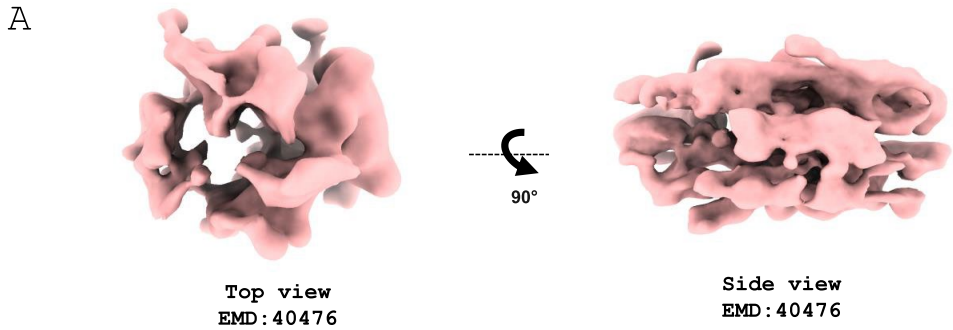

Fig. 5 Cryo-ET reconstructions revealing dynamic conformational states inaccessible to AlphaFold predictions. **A Human cardiac thick filament (EMD-40476):** top and side views display distinct myosin head conformations within the sarcomere, highlighting the regulatory role of cardiac myosin-binding protein C (cMyBP-C) during contraction. These conformational states, resolved in situ by cryo-ET, illustrate the conformational plasticity and state transitions that remain beyond the scope of AlphaFold's static structural predictions

and offered insights into regions not amenable to crystallography. Yet the study explicitly stated that *"comprehensive crystal structures were determined"*, underlining that AlphaFold served a supplementary role rather than being the primary method. Predictions mainly acted to *"validate experimental structures"*, demonstrating AlphaFold's role as a useful complement but not an independent solution (Fig. 6F).

From **EMD-19136 (SNO-GSTO1 study, also associated with EMD-43286)** [29], researchers investigated the structural basis of human S-nitrosylase in substrate recognition and catalysis using X-ray crystallography and cryo-EM. AlphaFold models offered *"excellent initial models"* and particular assistance for flexible regions and active sites involving conformational changes, filling in knowledge gaps where experimental data were incomplete. Nevertheless, its role was distinctly supplementary. The fundamental framework came from crystallography and cryo-EM, with AlphaFold acting as an assistive tool rather than solving the structure de novo.

3.5 Protein Engineering and Systems-Level Regulation

From **EMD-50672 (mScarlet-I protein engineering study)** [30], researchers focused on engineering the fluorescent protein for reduced nonspecific binding and enhanced performance. This study primarily relied on rational mutagenesis, spectroscopy, and microscopy. AlphaFold was not used for primary determination or design, underscoring a limitation: while powerful at predicting natural proteins, it does not extend to de novo design or predicting the outcomes of engineered mutations. Functional improvements required empirical validation through direct biochemical and biophysical characterization (Fig. 7A).

From **EMD-42509 (immune evasion in cancer study)** [31], researchers investigated transcriptional regulation and antigen presentation pathway defects within the tumor microenvironment. Although the core structures (e.g., MHC class I molecules) are well established, AlphaFold cannot capture conformational plasticity, post-translational modifications, or dynamic regulation networks that shape immune–tumor interactions. Cryo-EM visualizations were essential to uncover these mechanisms (Fig. 7B).

4 Summary

By reading across these studies, we can see that while AlphaFold has demonstrated significant advantages in predicting the structures of individual proteins or simpler complexes, its utility is inherently linked to its design. As a tool primarily for proteins, it cannot predict the structures of nucleic acids, lipids, or other non-protein macromolecules, nor simulate the interactions of these molecules within dynamic cellular environments.

These limitations are evident in studies of large in situ assemblies like CRP–C1 (Fig. 4A) and LRRK2–microtubules (Fig. 4B), higher-order chromatin organization (Fig. 4C–D), dynamic myosin head transitions (Fig. 5A), and complex functional systems such as protein engineering and immune evasion (Fig. 7).

Despite these boundaries, AlphaFold is widely and effectively used as a complementary tool: it verifies experimental results, supplements structural information (e.g., ZBTB20, Fig. 6A–E), predicts flexible or unresolved domains (e.g., MYPT3 ankyrin repeats, Fig. 6F), and offers initial models where data are incomplete (e.g., SNO-GSTO1). The benefits include completing structural models, guiding mutational analyses, and bridging gaps where experimental methods face challenges with dynamic or flexible regions. In these contexts, AlphaFold enhances and complements experimental workflows, but does not replace them.

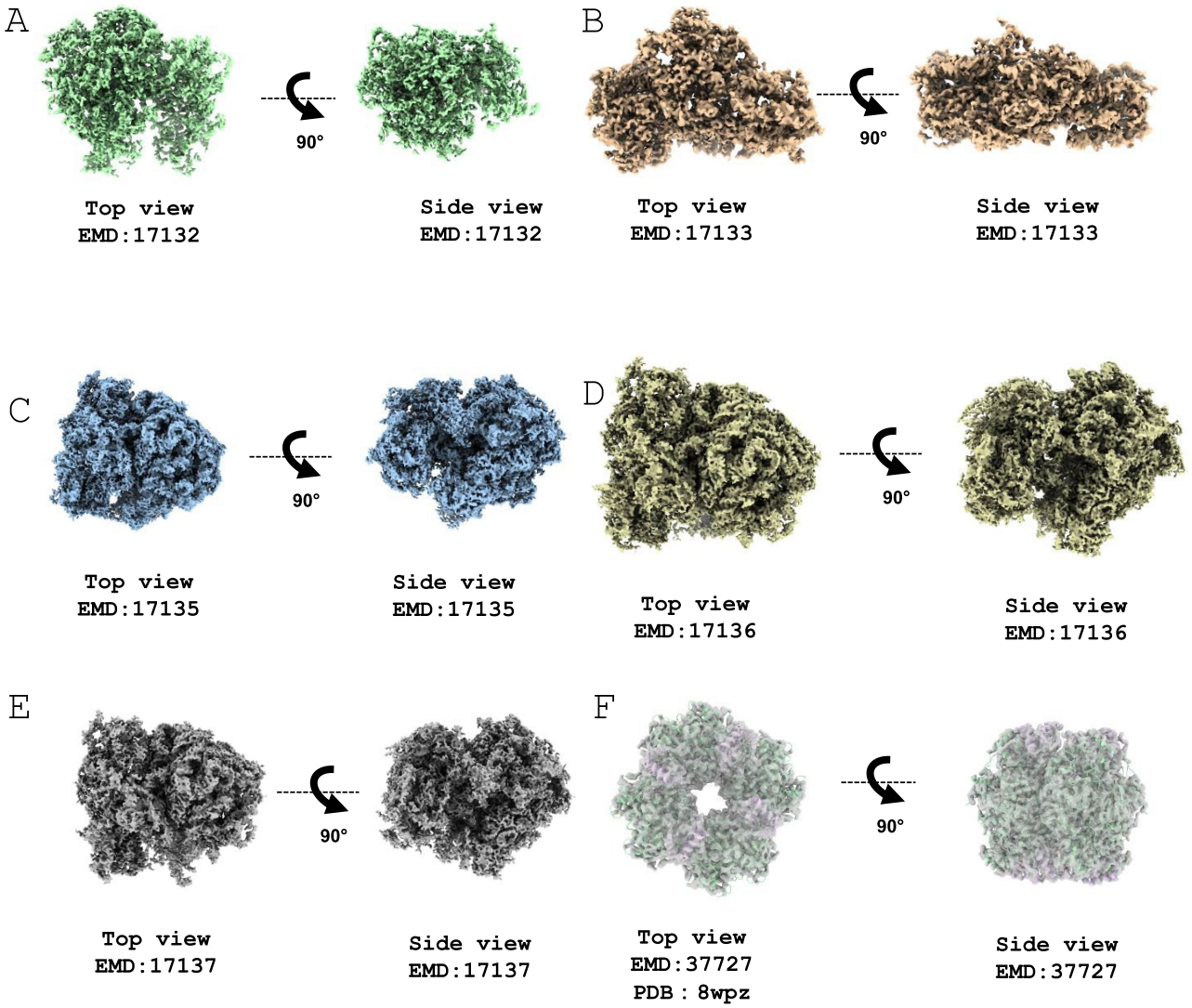

Fig. 6 Cryo-EM reconstructions illustrating AlphaFold's role as complementary starting models rather than standalone solutions. **A–E ZBTB20 transcriptional repression complex (EMD-17132, EMD-17133, EMD-17135, EMD-17136, EMD-17137):** cryo-EM and SAXS reconstructions revealed the architecture of ZBTB20 assemblies. AlphaFold predictions of the BTB domain provided high-confidence initial models for flexible or poorly resolved regions, but these required multiple rounds of refinement and validation against experimental density. The study highlights that AlphaFold served primarily as a computational hypothesis generator, assisting interpretation but not independently resolving the full assembly. **F Ppp1r12c/MYPT3 ankyrin repeat domains (EMD-37727; PDB: 8wpz):** cryo-EM and crystallography established definitive structures of the ankyrin repeats. AlphaFold produced "excellent initial models" and complemented experimental work by filling unresolved regions, but its predictions were secondary to crystal data and experimental validation. *Together, these studies demonstrate that AlphaFold is most effective as an auxiliary tool—providing starting models, filling structural gaps, and guiding mutagenesis—while comprehensive structural resolution requires integration with cryo-EM, SAXS, or crystallography*

5 Discussion

5.1 Understanding the Frontiers of AlphaFold's Predictive Prowess

While AlphaFold has indisputably transformed structural biology by providing near-experimental accuracy for a wide range of proteins, a complete understanding of its predictive power requires acknowledging its boundaries. These limitations arise not only from the scope of its training data but also from the complexity and dynamism of real biological systems, where structure and function are rarely static.

A key limitation is AlphaFold's reliance on large datasets of previously determined, predominantly soluble and well-folded proteins. This introduces a structural bias that limits performance for novel folds, membrane proteins, and multi-subunit assemblies lacking homologous templates. For example, chromatin and sarcomere assemblies (Figs. 4C–D

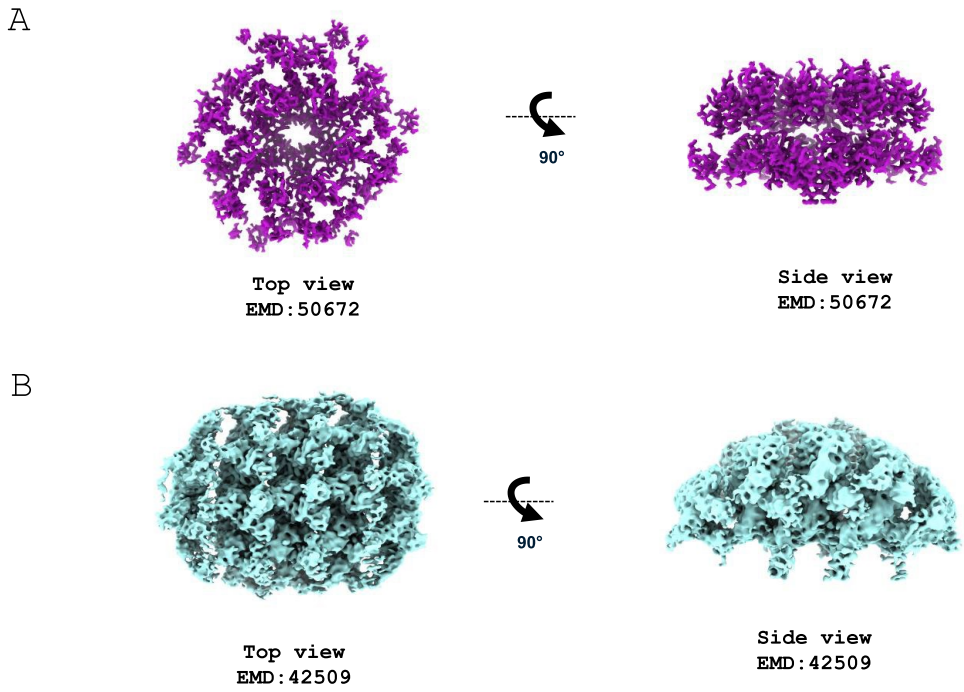

Fig. 7 Cryo-EM reconstructions illustrating functional protein engineering and systems-level regulation beyond the scope of AlphaFold predictions. **A Engineered mScarlet-I fluorescent protein (EMD-50672):** cryo-EM density maps show structural modifications introduced through rational mutagenesis to reduce nonspecific binding and enhance photophysical properties. Functional evaluation required empirical imaging and photophysical assays, underscoring AlphaFold's current limitation in predicting performance outcomes of engineered protein variants. **B Immune evasion mechanisms in cancer (EMD-42509):** cryo-EM reconstructions revealed molecular assemblies involved in antigen presentation pathway defects within the tumor microenvironment. Structural heterogeneity, post-translational modifications, and dynamic regulation observed in situ were not captured by AlphaFold's static predictions, emphasizing the necessity of experimental approaches for understanding immune–tumor interactions

and 5A) involve extended, flexible architectures whose modular interfaces are incompletely represented in AlphaFold's training distribution. These cases highlight how the method's data-driven priors constrain generalization beyond canonical globular folds.

AlphaFold's difficulties with intrinsically disordered proteins (IDPs) further illustrate the limits of its design. Proteins such as α-synuclein, Tau, and the p53 transactivation domain adopt heterogeneous, dynamic conformations in vivo but are returned by AlphaFold as static, fragmented, low-confidence predictions (pLDDT < 50). Experimental evidence from NMR, SAXS, and single-molecule fluorescence confirms that these regions exist as ensembles of interconverting structures, emphasizing that static predictions cannot capture their biological relevance.

Challenges also arise in protein–protein interaction (PPI) modeling. Although AlphaFold-Multimer extends AlphaFold's capacity to predict quaternary structures, it struggles with weak or transient interactions where interface geometries vary across conditions. For instance, in complexes such as Hsp90–cochaperone and ubiquitin–E2 enzyme assemblies, AlphaFold-Multimer often misassigns partner orientation or fails to model the flexible linker regions that mediate binding. These issues are particularly common in low-affinity or regulatory assemblies, where small energetic perturbations yield large conformational variability. Likewise, ligand-induced transitions and ternary PROTAC complexes (Fig. 5C) remain beyond the system's resolution because such rearrangements involve context-dependent energetics that cannot be inferred from sequence alone.

Another major challenge lies in transmembrane protein prediction. Unlike soluble proteins, membrane proteins are stabilized by lipid interactions and asymmetric electrostatic environments that are absent from AlphaFold's training data. As a result, helical bundle orientation, loop topology, and dynamic coupling to lipid bilayers are often mispredicted. Comparisons with cryo-EM reconstructions of GPCRs and ABC transporters show that AlphaFold2 correctly captures the general fold but mispositions helices by up to 3–5 Å in low-homology regions. Recent efforts such as MembraneFold and AF2Complex have begun to address these issues by embedding physical membrane constraints or coevolutionary interface priors into the prediction pipeline.

The modeling of large or in situ macromolecular assemblies also exposes AlphaFold's limitations. Complexes like CRP–C1 and LRRK2–microtubule filaments (Fig. 4A–B)

exhibit compositional heterogeneity and contextual flexibility that are only discernible under native cellular conditions. AlphaFold cannot capture such mesoscale organization, which depends on crowding, cofactors, and post-translational modifications. Similarly, multi-chain docking in AlphaFold-Multimer remains rigid; dynamic rearrangements and conformational gating are not yet captured, limiting its use for modeling transient assemblies and viral capsids.

To overcome these barriers, integrative workflows combining AI-based prediction with experimental validation are increasingly crucial. In the ZBTB20 and MYPT3 studies, AlphaFold2 models were refined using cryo-EM maps, SAXS profiles, and crystallographic restraints via tools such as Phenix real-space refinement and ISOLDE. This integration provided domain-level accuracy while correcting misoriented linkers and flexible loops. Similar strategies can be expanded to include cross-linking mass spectrometry, chemical labeling, and HDX–MS, which offer empirical distance or dynamics constraints to guide refinement. However, we also note that aligning predicted coordinates with experimental density can be non-trivial; discrepancies in scaling, overfitting, or partial density matching can propagate bias if not carefully validated.

Despite its computational efficiency relative to traditional physics-based simulations, AlphaFold remains resource-intensive for proteome-scale or multi-chain systems. GPU-based inference for large complexes still represent a bottleneck for smaller research groups, limiting widespread accessibility and benchmarking diversity.

Finally, the dynamic nature of proteins underscores the conceptual gap between AlphaFold's static predictions and biological reality. Proteins continuously sample conformational ensembles, undergo modifications, and participate in transient networks. Future development must integrate molecular dynamics, ensemble refinement, and machine-learning models trained on time-resolved data to move toward a dynamic, context-aware representation of protein structure.

We also acknowledge that our dataset of 32 cryo-EM and cryo-ET studies, while systematically selected, captures only a fraction of the literature available by 2024. Nevertheless, this focused corpus allowed us to extract quantitative insights into AlphaFold2 adoption patterns, identify recurring experimental contexts where it excels or fails, and highlight emerging integrative workflows—including its application to in situ cryo-ET, membrane assemblies, and plant-specific transcriptional complexes (Fig. 5A). These observations extend beyond previously published general reviews by linking specific structural contexts to predictive performance trends, thereby providing a comparative framework for the community to assess AlphaFold's evolving role in structural biology.

5.2 Limitations

This review has several limitations that should be acknowledged. First, our dataset is restricted to 32 cryo-EM and cryo-ET studies deposited in the EMDB in 2024, which provides a focused but non-exhaustive snapshot of current AlphaFold usage; therefore, trends identified here may not fully represent the broader structural biology community. Second, the classification of "AlphaFold contribution" relies on descriptions provided by the original articles, which vary in detail and may introduce reporting bias. Third, we did not include non-EM structural techniques such as NMR, X-ray crystallography, or XL-MS workflows, which may influence how AlphaFold is adopted across disciplines. Finally, our synthesis emphasizes qualitative interpretation rather than benchmarking predictive accuracy, and future meta-analyses incorporating quantitative metrics could further refine these observations.

5.3 Future Perspectives

The extraordinary success of AlphaFold2 has set a new standard in structural prediction, but significant opportunities remain to expand its scope, improve its accuracy, and enhance its integration with experimental data. Examples throughout this study (Figs. 4, 5, 6, 7) illuminate key frontiers for future development.

One major direction is modeling protein dynamics. AlphaFold's static predictions do not reflect the reality of conformational landscapes. Integrating molecular dynamics simulations—such as those based on Markov state models—could enable ensemble-based predictions that capture intermediate and allosteric states. This would be especially beneficial in large, flexible systems like those observed in Fig. 5A, where transient domain movements are central to function.

Improving interaction prediction is another urgent goal. AlphaFold-Multimer has limitations in modeling weak, transient, or flexible interfaces, as demonstrated in the PROTAC ternary complex case (Fig. 5C). Here, enhanced interface-specific modeling that combines deep learning with co-evolutionary coupling and physics-based scoring could improve structural accuracy for drug discovery applications.

Extending predictive capabilities to intrinsically disordered regions (IDRs) also represent a crucial challenge. Experimental techniques like NMR and single-molecule FRET could provide valuable restraints for machine learning models to generate conformational ensembles, especially

for signaling proteins or plant-specific IDPs (Fig. 5A). This would move beyond AlphaFold's current tendency to return collapsed or uncertain outputs for disordered segments.

Integrative modeling is likely to become the norm, rather than the exception. Combining AlphaFold predictions with experimental density maps, chemical crosslinking, or SAXS profiles—as in Fig. 6A–F—can help disambiguate flexible domains, validate domain orientation, and complete partial reconstructions. For example, cryo-EM data in the ZBTB20 complex (Fig. 6A–E) and the ankyrin repeat domains of MYPT3 (Fig. 6F) highlight how predictive models can serve as robust starting points, but require experimental anchoring to reach atomic precision.

In parallel, the ability to integrate ligand and small-molecule docking into predictive frameworks could greatly enhance AlphaFold's utility in pharmacological design. This is particularly pressing in contexts such as immune evasion (Fig. 7B), where small differences in binding geometry or glycosylation can profoundly affect antigen presentation.

Future structure prediction frameworks are expected to integrate generative diffusion models and multimodal datasets, combining cryo-EM density, NMR restraints, and sequence co-evolution to bridge the gap between static prediction and dynamic function.

Finally, accessibility and interpretability must improve. Future tools should offer intuitive interfaces, clear uncertainty metrics, and easy export to modeling pipelines. Visual aids that link structural predictions to biological function would empower experimentalists to generate more targeted hypotheses, and in turn, produce better training data for future AI models.

5.4 Toward an Integrative, Context-Aware Structural Biology

AlphaFold is not the endpoint—it is the launchpad. Its greatest impact will emerge not from replacing traditional methods, but from elevating them into a more predictive, integrative paradigm. From resolving membrane-bound CRP platforms (Fig. 4A) to refining conformationally heterogeneous chromatin structures (Fig. 4C–D), and from stabilizing dynamic sarcomere assemblies (Fig. 5A) to accelerating structure-based mutagenesis (Fig. 6), the synergy between AI and experimentation is unmistakable.

Moving forward, structural biology will depend increasingly on hybrid workflows that combine high-accuracy AI predictions with rich experimental context. Together, these tools promise not only to solve structures—but to understand how they move, interact, and function inside living systems.

6 Conclusion

This study presents a systematic and comparative analysis of 32 cryo-EM and cryo-ET studies published in 2024, providing quantitative evidence for the growing integration of AlphaFold into structural biology. Among these, 25 studies (78.1%) employed AlphaFold2 in some capacity, while 7 did not. In our analysis, "contribution" was operationally defined as AlphaFold being used for at least one of the following: (i) model completion in unresolved or low-density regions, (ii) domain or subunit fitting within cryo-EM maps, or (iii) generation of structure-based functional hypotheses (e.g., mutagenesis targets or conformational interpretation). Under this definition, AlphaFold enabled atomic-level modeling (pLDDT > 90) in the majority of AlphaFold-utilizing cases, including complex systems such as viral core assemblies (EMD-17704) [13], chromatin remodelers (EMD-37966) [12], and E3 ligase–substrate complexes (EMD-38163) [11].

AlphaFold's utility was most evident where experimental resolution or local map quality limited de novo building, enabling structure-based hypotheses, mutagenesis guidance, and subunit assignment. Its synergy with cryo-EM supported structural elucidation even in challenging contexts, exemplified by rabies virus matrix proteins (EMD-46612) [19] and host–pathogen interfaces such as the SARS-CoV-2 ORF6 nuclear pore blockade (EMD-19137) [17]. At the same time, our dataset also exposes clear boundaries. In CRP–C1 complexes (EMD-18830) [32] and LRRK2–microtubule assemblies (EMD-40525) [23], AlphaFold failed to recapitulate domain movements, flexible linkers, and compositional heterogeneity evident in cryo-ET maps—illustrating that high local confidence does not guarantee biological completeness in native, heterogeneous environments. Likewise, mitotic chromatin compaction (EMD-37002; EMD-19459) [24, 25] underscores mesoscale packing and cell-state transitions that remain experiment-resolved rather than prediction-resolved.

Beyond restating known limitations (e.g., flexible or disordered regions), this review adds new, comparative insights from the 2024 corpus: (1) adoption patterns—AlphaFold2 was applied most frequently to viral and bacterial assemblies typically mapped at ~ 7–20 Å, while higher-resolution eukaryotic complexes showed reduced dependence; and (2) mode of use—across AlphaFold-utilizing studies, most instances fell into domain-level completion/validation rather than full complex determination, aligning with the method's current strengths. These meta-level observations help contextualize when AlphaFold materially improves outcomes versus when orthogonal data are decisive.

Looking forward, integrative workflows will define the next phase of discovery. In practical terms, this means

iterative fitting and refinement that couples AlphaFold starting models with density-guided rebuilding and validation (e.g., real-space refinement, interactive flexible refinement, cross-validation with biochemical restraints), alongside MD-based ensemble sampling to probe conformational variability. Such pipelines are already enabling reweighting by density correlation, confidence-aware editing of low-pLDDT segments, and targeted experimental design, thereby moving from static predictions toward dynamic, context-aware structural hypotheses. In this view, the convergence of computation and experiment is not replacing bench validation but redefining its scope—accelerating the path from structure determination to functional interpretation and rational design.

Finally, we explicitly acknowledge that our dataset—32 EMDB-linked studies from a single year—is a focused, illustrative snapshot rather than an exhaustive survey. Nonetheless, to our knowledge, this is the first systematic, quantitative synthesis of AlphaFold utilization specifically within EMDB-deposited cryo-EM and cryo-ET structures in 2024. By coupling case-level evidence with cross-study statistics, the review delineates where predictive models align—or diverge—from biological reality and highlights priority areas where integrative modeling can most effectively advance structural biology.

6.1 EMDB-Based Dataset Collection and Literature Screening

We performed a time-bounded survey of the Electron Microscopy Data Bank (EMDB), which archives 3D cryo-EM and cryo-ET maps/tomograms and associated metadata (formal resource citation provided in the References). To capture current practice, we restricted the search to entries deposited between 1 January and 31 December 2024. The EMDB query returned all 2024 deposits (N = <N_total_2024>; see the legend of Table 1), from which we curated a study set of 32 entries with peer-reviewed primary articles suitable for AlphaFold usage assessment (Table 1; Supplementary Fig. S1).

Topic filters and rationale. To ensure biological and methodological diversity while keeping the review tractable, we used stratified purposive sampling across organismal contexts and resolution regimes. Specifically, we defined topic bins—Homo sapiens, Sus scrofa, vaccinia virus/viral assemblies, other viral vaccines, and a resolution-focused bin—to (i) represent both eukaryotic and viral systems that dominate 2024 depositions, (ii) balance single-particle cryo-EM and in situ cryo-ET exemplars, and (iii) include a near-atomic subset for contrastive analysis. Counts per bin (e.g., Sus scrofa, n = 6; vaccinia virus, n = 5; etc.) are reported in Table 1. Following community convention, we classify high resolution as < 3.0 Å, and 3.0–4.0 Å as near-atomic/medium–high; the latter category (9 entries) is labeled accordingly in the tables to avoid ambiguity.

Inclusion/exclusion criteria. We included EMDB entries that (i) had a peer-reviewed primary publication in 2024 linked to the deposition; (ii) reported experimental cryo-EM or cryo-ET data as the basis for structural interpretation; and (iii) provided sufficient methodological detail to verify whether AlphaFold/AlphaFold2 (monomer) and/or AlphaFold-Multimer were used. We excluded purely computational papers without experimental maps, preprints lacking peer review for the primary structure report, and entries missing sufficient methodological detail. The screening flow and counts at each stage (initial retrieval, screened, included/excluded with reasons) are summarized in Supplementary Fig. S1 and Supplementary Table S1.

Ascertainment of AlphaFold usage. For each included study, two reviewers independently performed full-text manual screening supplemented by keyword searches ("AlphaFold", "AlphaFold2", "AF2", "Multimer", "ColabFold", "predicted model", "fitted AF2 domain"). Disagreements were resolved by consensus; Cohen's κ for the binary label "AF used vs not used" is reported in Supplementary Table S1. Where AlphaFold was used, we coded mode of use into three non-exclusive categories: initial model completion, domain/subunit fitting/validation, and functional inference (e.g., guiding mutagenesis). These category counts are reported in Table 1.

6.2 Data Extraction, Coding, and Quantitative Synthesis

For each study we extracted: organism/system, technique (cryo-EM/cryo-ET), nominal and local resolution (Å), EMDB/PDB accessions, AlphaFold version (monomer vs Multimer), presence of pLDDT/PAE confidence reporting, and the use-mode categories above. Data were tabulated in a version-controlled spreadsheet (Supplementary Data File 1) and analyzed in R/Python (scripts provided as Supplementary Code). Summary statistics (proportions and 95% CIs) and association tests (χ^2/Fisher's; Spearman's ρ for resolution vs usage) are presented in the Results and Supplementary Fig. S1.

6.3 Structural Visualization and Fit Assessment

EMDB maps (.map/.mrc) were downloaded via the EMDB web interface and programmatically using the EMDB REST API for batch retrieval (accession list in Supplementary Data File 1). Visualizations were prepared primarily in UCSF ChimeraX (v<ChimeraX_version>) with confirmation in UCSF Chimera (v<Chimera_version>). Maps were

displayed with Volume Viewer, and atomic models (PDB or deposited models) were fitted using Fit in Map. Where AlphaFold predictions were integrated by the original authors, we overlaid the corresponding AlphaFold models and annotated pLDDT/PAE where available.

Contour-level policy. We adopted the EMDB-recommended contour as the starting level and adjusted within ± 10–20% to optimize visualization of contiguous signal while avoiding noise inflation; the chosen contour (absolute value or multiplier of the recommended level) is reported in figure legends. To assess agreement between model and density we used map–model correlation (ChimeraX Map–Model Fit, reporting CC_mask/CC_box) when models were available; qualitative overlays are used otherwise, with representative views captured from multiple angles. Where relevant, we additionally inspected local resolution maps to ensure that fitting claims did not rely on sub-threshold regions.

6.4 Definitions and Versioning

Throughout, "AlphaFold2" refers to the monomeric prediction framework; "AlphaFold-Multimer" denotes the multimeric complex mode; and "ColabFold" refers to community deployments of AF2/Multimer. When a paper stated "AlphaFold" generically, we resolved the version from the text, methods, or repository metadata; unresolved cases are labeled "AF (unspecified)" in Table 1.

6.5 Transparency, Reproducibility, and Limitations

All screening criteria, coded variables, accession lists, and analysis scripts are provided in Supplementary Data File 1 and Supplementary Code. The curated dataset size is 32 studies (25 with AlphaFold usage; 7 without), as summarized in Table 1 with system-level breakdowns. We note that restricting to one calendar year yields a focused but non-exhaustive corpus; accordingly, our statistical syntheses are descriptive, and inferences are framed with this scope in mind.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10930-025-10310-8.

Acknowledgements We gratefully acknowledge Dr. Guodong Xie (Postdoctoral Researcher, University of Chicago) for his valuable technical advice on cryo-EM map processing and model fitting. His feedback significantly improved our methodology and analysis.

Authors Contributions G.H., M.Q.C.L., and S.-R.L. contributed equally to this work. G.H. led the writing of high-weight manuscript sections, including major portions of the Results and Introduction, and coordinated cross-section integration. M.Q.C.L. co-led the Results sections, contributed significant data interpretation, and achieved the highest classroom performance. S.-R.L. integrated multi-author contributions, wrote technically complex Results sections, and demonstrated strong engagement throughout. S.W. contributed to the Introduction and Results sections and drafted the initial version of the Conclusion. L.E.N.T. contributed to the Results section. J.Z. supervised the entire project, provided methodological guidance, and finalized the manuscript. All authors participated in discussions, figure preparation, and manuscript revision.

Data Availability No datasets were generated or analysed during the current study.

Conflict of interest The authors declare that they have no conflict of interest.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process During the preparation of this work, the author(s) used **ChatGPT (OpenAI, GPT-4 model)** to assist with **language refinement, formatting consistency, figure legend editing, and summarizing technical content**. After using this tool, the author(s) carefully reviewed, edited, and validated all content to ensure accuracy and clarity. The author(s) take full responsibility for the integrity and originality of the content presented in this publication.

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References

1. Nogales E, Scheres SHW (2015) Cryo-EM: a unique tool for the visualization of macromolecular complexity. Mol Cell 58(4):677–689. https://doi.org/10.1016/j.molcel.2015.02.019
2. Fromm SA, Sachse C (2016) Cryo-EM structure determination using segmented helical image reconstruction. In: Crowther RA (ed) The resolution revolution: recent advances in cryo-EM. Methods Enzymol, 579:307–328. https://doi.org/10.1016/bs.mie.2016.05.034
3. Merk A, Bartesaghi A, Banerjee S et al (2016) Breaking cryo-EM resolution barriers to facilitate drug discovery. Cell 165(7):1698–1707. https://doi.org/10.1016/j.cell.2016.05.040
4. Orlova EV, Saibil HR (2011) Structural analysis of macromolecular assemblies by electron microscopy. Chem Rev 111(12):7710–7748. https://doi.org/10.1021/cr100353t
5. Cheng Y, Grigorieff N, Penczek PA, Walz T (2015) A primer to single-particle cryo-electron microscopy. Cell 161(3):438–449. https://doi.org/10.1016/j.cell.2015.03.050
6. Doerr A (2017) Cryo-electron tomography. Nat Methods 14(1):34. https://doi.org/10.1038/nmeth.4115
7. Hagen WJH, Wan W, Briggs JAG (2017) Implementation of a cryo-electron tomography tilt-scheme optimized for high

resolution subtomogram averaging. J Struct Biol 197(2):191–198. https://doi.org/10.1016/j.jsb.2016.06.007
8. Yang Z, Zeng X, Zhao Y, Chen R (2023) AlphaFold2 and its applications in the fields of biology and medicine. Signal Transduct Target Ther 8:115. https://doi.org/10.1038/s41392-023-01381-z
9. Mirabello C, Wallner B, Nystedt B, Azinas S, Carroni M (2024) Unmasking AlphaFold to integrate experiments and predictions in multimeric complexes. Nat Commun 15(1):8724. https://doi.org/10.1038/s41467-024-52951-w
10. Jumper J, Evans R, Pritzel A et al (2021) Highly accurate protein structure prediction with AlphaFold. Nature 596(7873):583–589. https://doi.org/10.1038/s41586-021-03819-2
11. Chen M, Chen X, Zhao J et al (2025) Molecular architecture of mammalian pyruvate dehydrogenase complex. Protein Cell 16(1):72–78. https://doi.org/10.1093/procel/pwae044
12. Zhang Y, Su X, Li J et al (2024) Molecular architecture of the mammalian 2-oxoglutarate dehydrogenase complex. Nat Commun 15(1):8407. https://doi.org/10.1038/s41467-024-52792-7
13. Liu J, Wang X, Li Q et al (2024) The palisade layer of the poxvirus core is composed of flexible A10 trimers. Nat Struct Mol Biol 31(7):1105–1113. https://doi.org/10.1038/s41594-024-01218-5
14. Weismehl M, Schulze WM, Kainrath S et al (2024) Structural insights into the activation mechanism of antimicrobial GBP1. EMBO J 43(4):615–636. https://doi.org/10.1038/s44318-023-00023-y
15. Hernandez-Gonzalez M, Calcraft T, Nans A, Rosenthal PB, Way M (2024) Palisade structure in intact vaccinia virions. MBio 15(2):e03134-e3223. https://doi.org/10.1128/mbio.03134-23
16. Ma X, Li Y, Zhang H et al (2023) Palmitoylation at residue C221 of Japanese encephalitis virus NS2A protein contributes to viral replication efficiency and virulence. J Virol 97(6):e00382-e423. https://doi.org/10.1128/jvi.00382-23
17. Hoffmann PC, Kalveram L, Roberts A et al (2025) Nuclear pore permeability and fluid flow are modulated by its dilation state. Mol Cell 85(3):537-554.e11. https://doi.org/10.1016/j.molcel.2024.11.038
18. Cao D, Ma B, Cao Z, Zhang X, Xiang Y (2023) Structure of Semliki Forest virus in complex with its receptor VLDLR. Cell 186(10):2208-2218.e15. https://doi.org/10.1016/j.cell.2023.03.032
19. Cai X, Liu Y, Zhang Y et al (2024) Structural heterogeneity of the rabies virus virion. Viruses 16(9):1447. https://doi.org/10.3390/v16091447
20. Wang Y, Li X, Zhao J et al (2023) The RNA m6A landscape of mouse oocytes and preimplantation embryos. Nat Struct Mol Biol 30(5):703–709. https://doi.org/10.1038/s41594-023-00969-x
21. Dick A, Smith J, Lee H et al (2024) Structural characterization of Thogoto virus nucleoprotein provides insights into viral RNA encapsidation and RNP assembly. Structure 32(8):1068-1078.e5. https://doi.org/10.1016/j.str.2024.04.016
22. Noone DP, Smith JD, Brown M et al (2024) Structural basis for surface activation of the classical complement cascade by the short pentraxin C-reactive protein. Proc Natl Acad Sci 121(37):e2404542121. https://doi.org/10.1073/pnas.2404542121
23. Sheng K, Li N, Rabuck-Gibbons JN, Dong X, Lyumkis D, Williamson JR (2023) Assembly landscape for the bacterial large ribosomal subunit. Nat Commun 14(1):5220. https://doi.org/10.1038/s41467-023-40859-w
24. Chen JK, Lee S, Park J et al (2025) Nanoscale analysis of human G1 and metaphase chromatin in situ. EMBO J 44(9):2658–2694. https://doi.org/10.1038/s44318-025-00407-2
25. Lange F, Müller K, Richter T et al (2025) In situ architecture of the human prohibitin complex. Nat Cell Biol 27(4):633–640. https://doi.org/10.1038/s41556-025-01620-1
26. Chen L, Liu J, Rastegarpouyani H, Janssen PML, Pinto JR, Taylor KA (2024) Structure of mavacamten-free human cardiac thick filaments within the sarcomere by cryo-electron tomography. Proc Natl Acad Sci 121(9):e2311883121. https://doi.org/10.1073/pnas.2311883121
27. Xue L, Spahn CMT, Schacherl M, Mahamid J (2025) Structural insights into context-dependent inhibitory mechanisms of chloramphenicol in cells. Nat Struct Mol Biol 32(2):257–267. https://doi.org/10.1038/s41594-024-01441-0
28. Kong W-W, Zhang L, Li P et al (2024) Cryo-electron tomography reveals the packaging pattern of RuBisCOs in Synechococcus β-carboxysome. Structure 32(8):1110-1120.e4. https://doi.org/10.1016/j.str.2024.05.007
29. Tuijtel MW, Koster AJ, Hoogenboom JP et al (2024) Thinner is not always better: Optimizing cryo-lamellae for subtomogram averaging. Sci Adv 10(17):eadk6285. https://doi.org/10.1126/sciadv.adk6285
30. Burt A, Himes BA, Scheres SHW et al (2024) An image processing pipeline for electron cryo-tomography in RELION-5. FEBS Open Bio 14(11):1788–1804. https://doi.org/10.1002/2211-5463.13873
31. Electron Microscopy Data Bank (2025) Electron Microscopy Data Bank. https://www.ebi.ac.uk/emdb/EMD-42509. Accessed 1 Sept 2025
32. Gao Y, Yang H, Zhang L, Liu T, Huang W et al (2024) Structural basis of human P4HA2 collagen prolyl-4-hydroxylase assembly revealed by cryo-EM and AlphaFold2. Nat Commun 15:2671. https://doi.org/10.1038/s41467-024-20176-3

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